

Economies of agglomeration on fast growth companies

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Introduction

Background

It is not surprising to see in many places around the world, companies and small businesses agglomerated in one district or arounds. This agglomeration contributes significantly on decreasing price of production, deployment as it does on creating a more innovative and productive environment for their workers. Many of these effects are widely known on business clusters and lately have shown amazing results on other sectors such as biotech, hightech and so on, having a great example on BioPharma clusters in Switzerland or HighTech ecosystems in other countries.

Interest

The main goal of this project is to locate through Google API the companies listed on EMEA Fast Growth 500 report and check if they are located in any of these clusters of the cities or are outside of them. Once located we will see which venues we can find around their centroids and if there is any relation.

The success of a company is not determined by the location of the office although it can affect directly as these clusters do. Knowing that this project is a beginner-intermediate one and the subject chosen is complex for my level, the results will be shown as probable without determining the rest of the factors that can affect the success in the location of your company or your services in the arounds.

In any case this project will be improved over the time and I encourage anyone with more knowledge and interest in this field to use my work to find out more about the relation between business, biopharma or startup clusters and their location and services around the same. I think that these models can be of remarkable interest for all the countries that are developing their own clusters in order to do it successfully.

Data acquisition and cleaning

The data used is the EMEA Fast Growth 500 annual report of Deloitte and the location of the business clusters like the London one. We use the Foursquare API to locate the services around the companies and the google API to get the coordinates to be mapped.

The PDF has been converted to csv and cleaned of NaN and other unuseful data such as the number of pages, titles and others. Having the DataFrame created we select the countries and the industries sectors with more companies in order to get relevant data. As the report just shows the rate growth, the name of the company, country and industry sector, using the geocoder API of google is a must due to the fact that it can recognise the name of the companies without the address.

	Ranking	Company name for communication	Country	Growth	Industry sector
0	1	Revolut	UK	39754	Fintech
1	2	OakNorth	UK	30706	Fintech
2	3	Cloud&Heat Technologies GmbH	DE	21474	Environmental Technology
3	4	DivideBuy	UK	19572	Fintech
4	5	FINEWAY	DE	16594	Software
...	...	...	...	...	...
571	496	shopware AG	DE	159	Software
572	497	metacrew group GmbH	DE	159	Media and Entertainment
573	498	LESCAV	Belgium	158	Hardware
574	499	Paysera LT	Lithuania	158	Fintech
575	500	Arcus Global	UK	157	Software

After we get the location the DataFrame has the address and the coordinates. This procedure will be done for the case of Great Britain and Netherlands due to the fact that are the countries that have more companies on the list and therefore, the ones that provide relevant data to analyse.

Company name for communication	Country	Growth	Industry sector	loc	point	lat	lon
Revolut	UK	39754	Fintech	(7 Westferry Circus, Canary Wharf, London E14 ...	(51.5062984, -0.0258565, 0.0)	51.506298	-0.025857
OakNorth	UK	30706	Fintech	(6th Floor, Nightingale House, 65 Curzon St, M...	(51.507971, -0.1447259, 0.0)	51.507971	-0.144726
DivideBuy	UK	19572	Fintech	(First Floor, Brunswick Court, Brunswick St, N...	(53.0124212, -2.2207889, 0.0)	53.012421	-2.220789
Moteefe	UK	7394	Media and Entertainment	(84 Eccleston Square, Pimlico, London SW1V 1NP...	(51.4931108, -0.1435609, 0.0)	51.493111	-0.143561
Clearmatics	UK	4900	Software	(30 City Rd, London EC1Y 2AY, UK, (51.5226132,...	(51.5226132, -0.08738939999999999, 0.0)	51.522613	-0.087389
...	...	...	...	...	...	...	...
Cloudhouse	UK	183	Software	(Ugli Campus Block A, 56 Wood Ln, Shepherd's B...	(51.511679, -0.2235788, 0.0)	51.511679	-0.223579
Zappi	UK	180	Software	(Theatre House, 97 - 99 Camden High St, London...	(51.536677, -0.1409407, 0.0)	51.536677	-0.140941

Methodology

The stages from acquisition to deploy a solution have been the next:

- Acquisition of Data through Deloitte's PDF EMEA Fast Growth 2019 and the consequent cleaning.
- Location of the address, latitude and longitude of the companies listed using GeoCode API of Google.
- Mapping using folium the companies by the industry sector and cluster them with k-means.
- Identify services around companies by sector using Foursquare API.
- Analysing the results.

With the dataframes having the address and coordinates we first use folium to visualize the localization of them by sectors. This first step has been relevant to detect those companies that are way far from the other ones and then be a problem when we do the clustering, specially when we need the location of the centroid to send it to FourSquare API Location.

These DataFrames once with the companies outliers deleted, allow us to use the agglomerative clustering function which will cluster the companies, having the location as the main measure. We can see that six clusters give us a good distribution for Great Britain and seven for Netherlands.





Even though agglomerative clustering functions provide us with an easy and fast solution to see the distribution, we can not use any library to get the centroids. At this point is when K-means is so necessary. To get the coordinates the more accurately possible we introduce the number of clusters that we are working with and $n_{init} = 25$ in order to get an accurate approximation of that centroid.

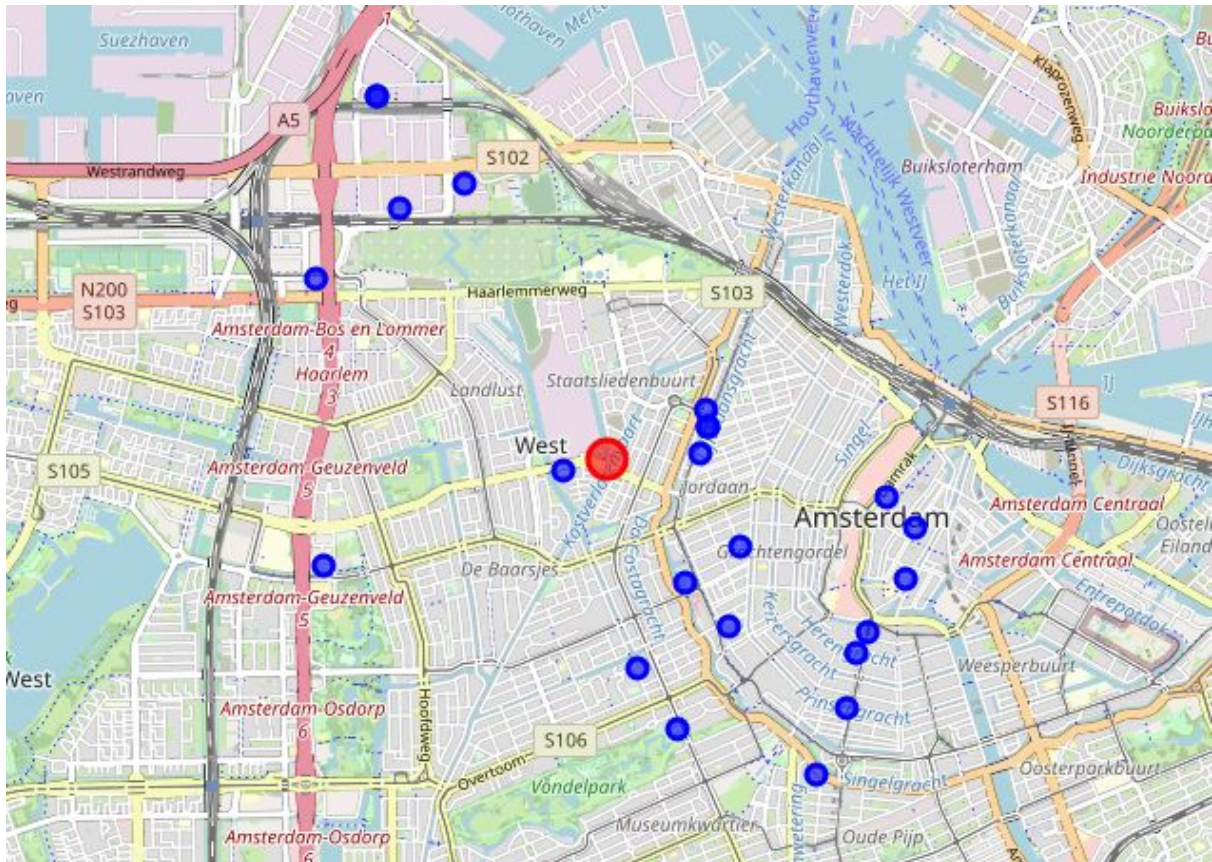
K-means has helped us to get the centroid of all those clusters in both countries but not all the clusters are interesting in the sense of their distribution, as is impossible to get a centroid that is relevant in a posterior analysis with FourSquare for his remote location.

Knowing that limitation, we use the centroids of London, Cambridge and Amsterdam.

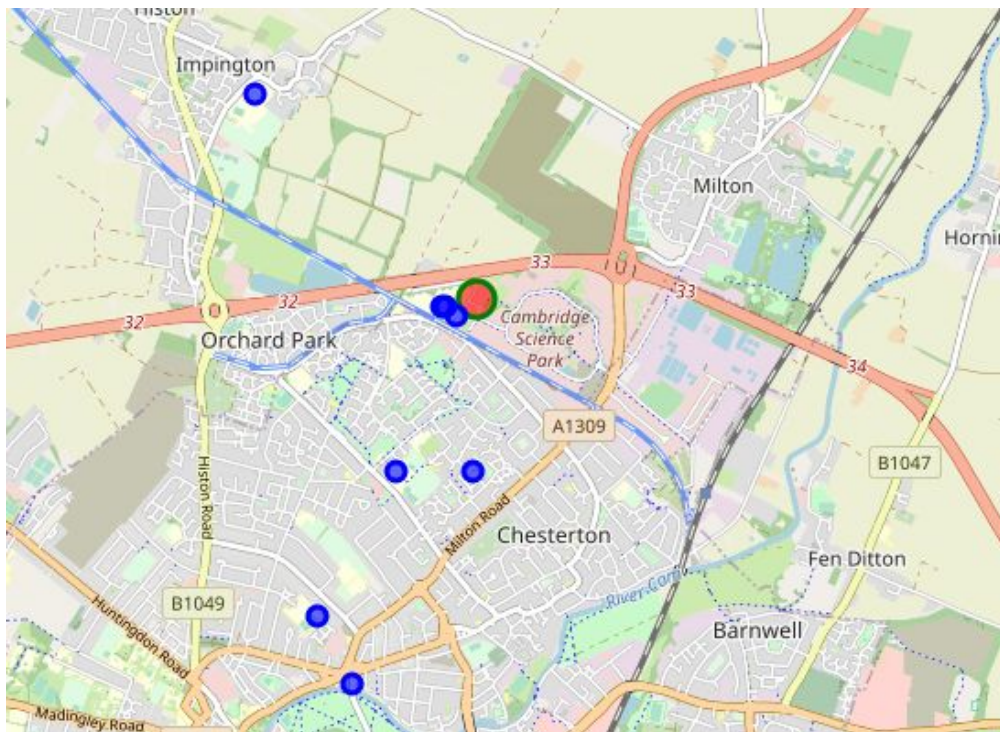
One of the multiple venues that we can analyse to see the repercussion on these clusters is the query of FourSquare 'Colleges & Universities' to see how related is a successful company with an academic environment. We know due to the previous analysis that 48 and 24 companies are in the London and Amsterdam clusters respectively but just four are in the Cambridge one.

It is important to say that the growth factor or the industry sector has not been used for this analysis as finding correlation using two API's that work with coordinates give us problems and we do not have the initial value of the company before this reached the financial peak, neither we know how the impact on sectors depending on countries in matters of laws, helps for development and research and so on.

Visualizing the results for the venues (Blue) in London, Cambridge and Amsterdam clusters we get the next maps having as centroids the red dots:



Amsterdam Venues in a radius of 2km



Cambridge venues in a radius of 2km



London venues in a radius of 2km

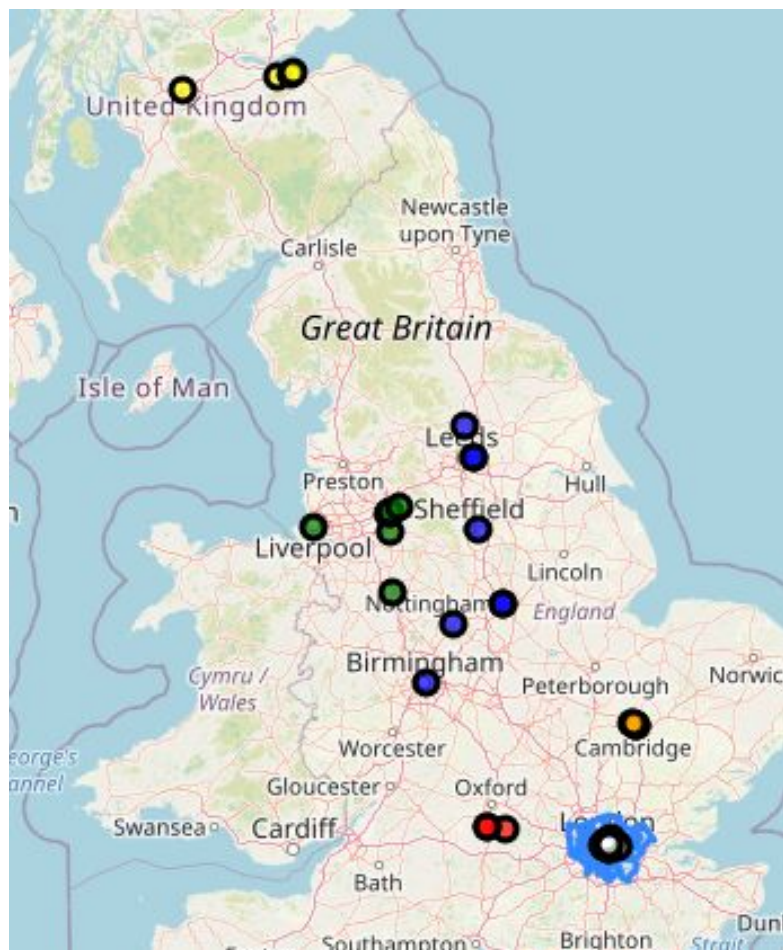
It does not take much time to see the clear difference between the cities. In the case of Cambridge we have seven venues within the radius related with colleges & universities for four companies, in Amsterdam we have twenty two venues within the radius for a total of twenty four companies and London have more than fifty venues within the radius for forty eight companies. To make an inferential analysis is not useful in this case due to the fact that we do not have enough clusters with enough companies to do so and the case of London has more than fifty venues which is the maximum number we can get from FourSquare API for free.

Even though we can't affirm the correlations backed on a statistical analysis it seems to be a relation. To determine these correlations is the key point for any data science project but the topic chosen has been way more complex to study than initially thought.

Discussion

After all the process is done, we can determine visually and just visually as pointed before, a correlation between the location and the success of that company. Through the agglomerative clusters of GB, we can appreciate clearly the distribution and the nucleus of these. In Scotland the cluster is defined with Glasgow and Edinburgh, in London the agglomeration is crystal clear, separately but with a similar distribution we have two clusters more, in Cambridge and Oxford respectively and then we have two clusters with an interesting composition.

We can see with less agglomeration, the nucleus in Manchester but having companies in Liverpool too and last but not least we see a distribution following almost vertically the cities of Leeds, Sheffield, Nottingham, Derby and Birmingham. In the last cluster it would be really interesting to check the communication facilities in the arounds because it seems to follow another sort of pattern than the other clusters with a relative proximity between companies in the same cluster. On the map it seems clearly communicated by a central highway.



Conclusion

In an exercise of self criticism I would say that I tried to extract conclusions of a subject that is for more experienced programmers and data scientists, as I spend barely two month by now learning Python. The remarkable interest of this subject is anyway worth trying because many times we see an extremely close scope on our disciplines and we totally lose the perspective of the problems to solve. In this case, we can try hundreds of ways to determine why a company is successful, checking their market cap, revenues and as many financial terms as we want, but seeing a relation in what we would not spend time thinking about as the location of these companies is sometimes awakening. It has been spending time researching for and interesting topic that I have discovered what it is the Economy of agglomerations and the benefits of this discipline.